
Label Swapping for Imbalanced Music Datasets

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Abstract

Data imbalance is a pervasive issue in machine learning that often hinders classifier performance by biasing predictions towards the majority class. Label-Noise-based Re-balancing (LNR) is a novel method proposed by Hu et al. (2025) that introduces beneficial label noise to adjust biased decision boundaries without the drawbacks of traditional resampling techniques. We apply this method to the imbalanced music dataset Meter2800 for time signature detection, meaning to improve the detection of rare, irregular meters (e.g., 5/4, 7/4) using a ResNet18 architecture. Our results show that beneficial label noise yields improved performance for recognizing minority class time signatures.

1 Introduction

Data imbalance is a common issue in machine learning that hinders classifier performance, leading to biased predictions favoring the majority class. In imbalanced datasets (e.g., 90% class A, 10% class B), accuracy is maximized by favoring class A, pushing the decision boundary deep into the territory of class B.

This results in poor performance for the minority class. Additionally, techniques such as resampling often introduce information loss or overfitting. Finding a solution is relevant in scenarios where over-representation of one class over another may be problematic.

Recently, Hu et al. (2025) proposed a new approach, Label-Noise-based Re-balancing (LNR), suggesting that adding beneficial label noise has the potential to adjust biased decision boundaries and improve performance without the drawbacks of traditional resampling [1].

Our project applies this method to imbalanced music datasets, specifically time signature detection. Meter2800 is an imbalanced dataset of musical excerpts labeled with time signatures, where standard meters 4/4 and 3/4 are abundant, while irregular meters such as 5/4 and 7/4 are scarce [2].

Reproducing the results of ResNet18 on Meter2800 and augmenting the training process with LNR, we seek to establish a new way of detecting complex musical meters.

Code repository is available at https://github.com/Matdrox/dl_music_label_swapping

2 Related Work

This project is based on two separate papers. Its purpose is to combine the novel method introduced in the former paper with the architecture of the latter.

2.1 Label Noise Re-balancing (LNR)

Binary classification on imbalanced datasets often results in biased decision boundaries, causing misalignment between accuracy and metrics such as F1-score. LNR addresses this by introducing

asymmetric label noise, selectively flipping majority class samples to the minority class based on their similarity to minority class samples. Unlike symmetric noise, which affects all classes equally, LNR focuses on majority samples that are confusing (i.e., they have high posterior probabilities of looking like the minority class). This process pushes the decision boundary back toward the majority region, reclaiming space for the minority class without discarding any data.

2.2 Time Signature Detection

Audio Similarity Matrices (ASM) and Beat Similarity Matrices (BSM) have been used as early methods for meter detection relying on signal processing. More recently, deep learning has shown superior performance. Abimbola et al. (2024) [3] demonstrated that a ResNet18 architecture using Mel-frequency Cepstral Coefficients (MFCCs) as input outperformed traditional machine learning models such as SVM and KNN, as well as other deep neural networks like CNN and CRNN. However, the dataset used in the paper, Meter2800, is highly imbalanced, making it a suitable candidate for testing the limits of the LNR method.

3 Methods

The final model architecture combines ResNet18 for time signature detection with the LNR algorithm for handling data imbalance. Imperfect representation is a classical problem in music information retrieval, where some musical concepts are inherently rare. Time signatures such as 5/4 and 7/4 are uncommon in Western music, while 4/4 and 3/4 dominate. This leads to imbalanced datasets that bias classifiers toward the majority classes, resulting in poor detection of minority class time signatures.

3.1 ResNet18 for Time Signature Detection

Time signature detection is approached as a supervised image classification problem using the methodology instructed by Abimbola et al. (2024) [?]. Raw audio signals are pre-processed into visual representations and fed into a ResNet18 architecture adapted for single-channel audio features.

Audio files are converted into Mel-frequency Cepstral Coefficients (MFCCs), compactly representing the spectral properties of the audio signal. For each 30-second audio clip, 13 cepstral coefficients are extracted over 130 time frames. In order to maintain consistency, the input dimensions are fixed to 130×13 (time frames $\times$ frequency coefficients). Each input sample x_i is thus treated as a single-channel image tensor $x_i \in \mathbb{R}^{1 \times 130 \times 13}$.

The Resnet18 architecture is a deep residual network consisting of 18 layers. While originally designed for three-channel RGB images (e.g., ImageNet), we adapt it for our low-resolution, single-channel audio features. MFCC inputs are processed through a series of residual blocks, letting it learn representations of the musical meters. The final layer is a fully connected layer that maps the extracted features to the output classes corresponding to the time signatures 3/4, 4/4, 5/4, and 7/4.

The original paper implements cross-entropy loss for an Adam optimizer with a learning rate of 0.001 training for 40 epochs. We have decided to work within the LNR framework, which uses SGD with (HYPERPARAMETERS) **motivera varför vi tog LNR**

3.2 LNR Algorithm

In order to enforce the model’s attention toward the minority class, LNR proposes the concept of beneficial lying. Instead of safely assuming the majority class, flipping the label of ambiguous samples to the minority class causes the model to pay better attention to the nuances of the minority class.

Lying randomly would confuse the model. Therefore, beneficial noise was introduced by the implementation of the LNR algorithm as follows:

- Let X be the input feature space (audio spectrograms) and $Y = \{0, 1\}$ be the binary label space, where 0 represents the majority class and 1 the minority class.
- Let f_θ be the ResNet classifier parameterized by weights θ .

- $\hat{\eta}(x_i) = P(Y = 1|X = x_i)$ denotes the estimated posterior probability (confidence score) that sample x_i belongs to the minority class.

- $I_{MA} = \{i|y_i = 0\}$ is the set of indices of majority class samples in the current batch.

The LNR algorithm identifies confusing majority samples based on their posterior probabilities. Calculating the mean μ and standard deviation σ of these probabilities after passing them through the classifier's softmax layer, the standardized Z-score Z_i for each majority sample is computed as:

$$Z_i = \frac{\hat{\eta}(x_i) - \mu}{\sigma}$$

Flip-rate probability $\rho(i)$ is computed using a hyperbolic tangent function and a threshold hyperparameter t_{flip} :

$$\rho(x_i) = \max(\tanh(Z_i - t_{flip}), 0)$$

Labels are flipped stochastically based on $\rho(x_i)$, as only majority samples with sufficiently high confusion (i.e., $Z_i > t_{flip}$) should have a non-zero chance of being flipped to the minority class. The algorithm is given in the pseudocode below:

Algorithm 1 Flip labels in binary classification with noise model $\rho(x)$

Require: Feature X and labels Y , Classifier C_f

Require: Threshold t_{flip}

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 $Ind_{MA} \leftarrow \text{index}(Y = 0)$ 
 $\hat{\eta}, Z, \rho \leftarrow \text{zeroVector}(\text{size} = N)$ 
for  $i \in 1 \rightarrow \text{length}(Y)$  do
     $\hat{\eta}[i] \leftarrow C_f(X[i])$ 
end for
 $\mu \leftarrow \text{mean}(\hat{\eta}[Ind_{MA}])$ 
 $\sigma \leftarrow \text{std}(\hat{\eta}[Ind_{MA}])$ 
for each  $i_{MA}$  in  $Ind_{MA}$  do
     $Z[i_{MA}] \leftarrow \frac{\hat{\eta}[i_{MA}] - \mu}{\sigma}$ 
     $\rho[i_{MA}] \leftarrow \max(\tanh(Z[i_{MA}] - t_{flip}), 0)$ 
     $U \leftarrow \text{Bernoulli}(\rho[i_{MA}])$ 
    if  $U == 1$  then
         $Y[i_{MA}] \leftarrow 1$ 
    end if
end for
return  $Y$ 

```

4 Data

Three datasets are used in this project.

4.1 KEEL

For binary classification experiments replicating the original LNR paper, the same KEEL datasets are used. KEEL is a repository of 99 binary imbalanced datasets. From these, 32 specific datasets were selected by Hu et al. (2025) for their experiments [4].

Their chosen datasets match the following criteria:

- a) Baseline already achieves an F1-score over 90%, or all methods have an F1-score lower than 10%.
- b) Datasets with insufficient minority class sample size in the test dataset (less than 10). We picked out four of those selected datasets to replicate the results of: *abalone-17_vs_7-8-9-10*, *ecoli1*, *glass0* and *vehicle1*.

4.2 CIFAR-10

Multiclass reproduction required us to use CIFAR-10, a well-known image classification dataset consisting of 60,000 32x32 color images in 10 classes. CIFAR-100 was also used in the original paper, but it was not necessary for our reproduction.

4.3 Meter2800

Meter2800 is a time signature dataset provided by Harvard Dataverse. It contains four annotated CSV files of audio data segmented into the meter classes 3, 4, 5 and 7. The dataset also contains a total of 2800 audio files of 30 seconds each. It exhibits severe class imbalance, making it a perfect candidate for testing the LNR method. Majority classes 4/4 and 3/4 dominate the dataset with 1200 samples each, while minority classes 5/4 and 7/4 are underrepresented with only 200 samples each.

5 Experiments and Findings

5.1 Binary classification on KEEL datasets

Table 1: F1 score, G-mean score and AUC with CART classifier on KEEL datasets

F1 score				
Dataset	Baseline	LNR	SMOTE	CC
abalone[...]	0.27±0.09	0.33±0.07	0.23±0.05	0.17±0.04
ecoli1	0.73±0.07	0.75±0.05	0.74±0.06	0.71±0.07
glass0	0.69±0.07	0.69±0.07	0.70±0.08	0.66±0.07
vehicle1	0.50±0.05	0.53±0.05	0.54±0.05	0.52±0.05
G-mean score				
abalone[...]	0.52±0.1	0.64±0.09	0.68±0.08	0.74±0.06
ecoli1	0.83±0.05	0.86±0.04	0.85±0.04	0.84±0.04
glass0	0.77±0.05	0.77±0.05	0.78±0.06	0.75±0.05
vehicle1	0.65±0.04	0.68±0.04	0.68±0.04	0.68±0.04
AUC				
abalone[...]	0.63±0.05	0.70±0.06	0.71±0.06	0.75±0.06
ecoli1	0.83±0.05	0.86±0.04	0.85±0.04	0.84±0.04
glass0	0.78±0.05	0.78±0.05	0.78±0.05	0.75±0.05
vehicle1	0.67±0.03	0.68±0.04	0.69±0.03	0.68±0.04

Table 1 and table 2 show the results of binary classification. LNR seems to have been the most advantageous for the *abalone-17_vs_7-8-9-10* dataset, and specifically for its f1 score, which might be because of its relatively big imbalance ratio of 39.31 [?]. Otherwise, the results are not clearly in the favor of LNR. The results are similar to the original LNR paper [1] in the sense that LNR outperforms the other methods in the greatest number of scenarios, even though the margins are quite slim.

5.2 Multiclass classification on CIFAR-10

5.3 Multiclass classification on Meter2800 dataset

6 Challenges

The LNR paper covers a novel, yet fairly approachable method for imbalanced dataset classification. Algorithms in form of pseudocode are provided, and the method is easy to implement.

However, the paper is quite broad. In an attempt to present the benefits of the proposed LNR method, experiments were conducted using many different frameworks. This requires the reader to not only

Table 2: F1 score, G-mean score and AUC with MLP classifier on KEEL datasets

F1 score				
Dataset	Baseline	LNR	SMOTE	CC
abalone[...]	0.21±0.16	0.37±0.13	0.33±0.07	0.22±0.05
ecoli1	0.77±0.06	0.78±0.06	0.76±0.06	0.75±0.06
glass0	0.68±0.11	0.68±0.07	0.69±0.08	0.66±0.07
vehicle1	0.66±0.05	0.69±0.08	0.69±0.04	0.66±0.05
G-mean score				
abalone[...]	0.33±0.21	0.60±0.18	0.80±0.08	0.81±0.13
ecoli1	0.85±0.05	0.85±0.04	0.87±0.04	0.87±0.04
glass0	0.76±0.10	0.77±0.05	0.76±0.10	0.74±0.06
vehicle1	0.77±0.04	0.81±0.09	0.80±0.04	0.79±0.04
AUC				
abalone[...]	0.91±0.08	0.90±0.08	0.91±0.04	0.90±0.07
ecoli1	0.95±0.02	0.94±0.03	0.94±0.03	0.94±0.03
glass0	0.85±0.06	0.84±0.05	0.84±0.06	0.82±0.05
vehicle1	0.89±0.03	0.89±0.05	0.89±0.02	0.87±0.03

Table 3: Accuracy and ECE for different methods (BRYR VI OSS OM ECE?)

Method	Acc@1	Acc@5	H-Acc	M-Acc	T-Acc	ECE
LNR	83.69 ± 0.05	99.01 ± 0.02	86.29 ± 0.10	80.58 ± 0.05	85.26 ± 0.03	9.86 ± 0.07
MisLAS	83.73 ± 0.01	99.06 ± 0.01	86.43 ± 0.03	80.63 ± 0.02	85.14 ± 0.03	9.94 ± 0.01
SelMix	80.55 ± 0.19	96.87 ± 0.19	79.93 ± 0.43	80.12 ± 0.16	81.75 ± 0.35	14.52 ± 1.30

understand the theory behind the proposed method, but also the details of the various frameworks used in the experiments. This makes it difficult to reproduce the results without prior knowledge of the frameworks.

Additionally, the structure of the paper makes it difficult to follow the experimental setups. The datasets used are scattered throughout the paper, and it is not always clear which datasets were used for which experiments. This requires the reader to constantly refer back to previous (or future) sections to understand the context of the experiments. Even then, some details are left out, making it difficult to fully understand the experimental setups.

Upon a close read of the source code provided by the authors of the LNR paper, we found that the original implementation of the CIFAR-10 experiments lacked a distinct validation split and instead uses the test set for early stopping and model selection. A standard validation set is absent in older benchmarks such as the 2009 dataset CIFAR-10, but modern best practices recommend a separate validation set to avoid overfitting to the test set. We chose to replicate this behavior for means of reproduction, but fixed it for our Meter2800 experiments by creating a proper validation split.

Table 4: Macro F1-score for different t_{flip} on the Meter2800 dataset

t_{flip}	F1 100e	Acc 100e	T-acc 100e	F1 140e	Acc 140e	T-acc 140e
0	0.7643	81.83	68.0	0.7792	83.52	66.83
1	0.7645	81.86	67.5	0.7787	83.52	66.17
2	0.7655	81.91	67.5	0.7781	83.50	66.0
3	0.7644	81.88	67.17	0.7788	83.55	66.0
4	0.7644	81.88	67.17	0.7788	83.55	66.0
5	0.7638	81.86	67.0	0.7796	83.60	66.0
6	0.7638	81.86	67.0	0.7796	83.60	66.00
7	0.7638	81.86	67.0	0.7797	83.62	65.30

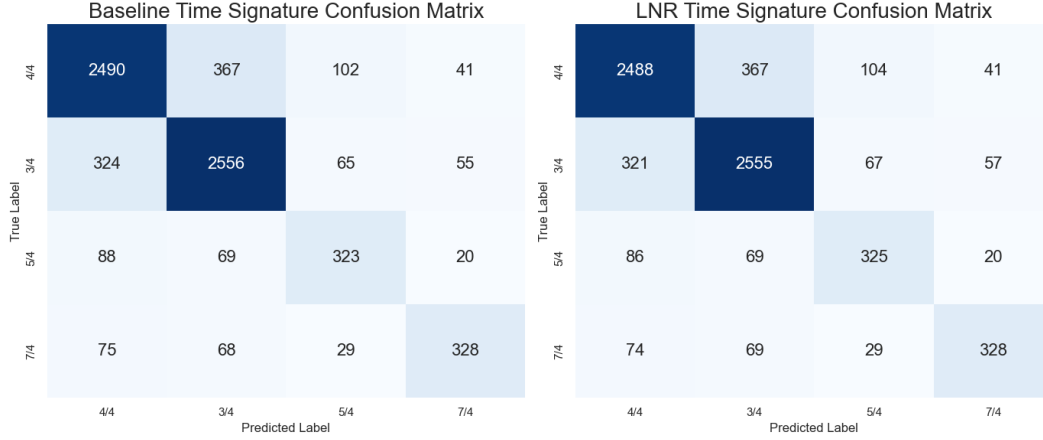


Figure 1: Confusion matrix for LNR on Meter2800 with $t_{flip} = 7$ after 140 epochs.

7 Conclusion

8 Ethical Consideration

Time signature detection is a niche task in music information retrieval with limited ethical implications. However, data integrity and representation remain important.

The Meter2800 dataset, while useful for its intended research purpose, may not fully represent the diversity of musical styles and cultures. This could lead to biased models that perform poorly on underrepresented time signatures or genres.

Fixing imbalanced datasets is an important task for machine learning in general. Real-world cases such as breast cancer detection often involve many more negative samples than positive ones, despite the importance of correctly identifying the positive cases. In these critical applications, deliberately mislabeling majority samples to improve sensitivity ought to be done with utmost care to avoid an unacceptable increase in false positives.

9 Limitations

Meter2800, although imbalanced and suitable for testing LNR, is relatively small with only 200 samples for the minority classes. The statistical significance of the improvement is harder to establish compared to large-scale image datasets like CIFAR-10.

10 Self Assessment

Assessing our project is done using the detailed assessment criteria provided by the course. The final result is a combination of reproduction and novel application of the LNR method to music time signature detection.

10.1 Grades E-D

This paper discusses the LNR method by describing its theory and implementation in detail. We argue why data imbalance is a problem and how LNR addresses it. The paper also reviews related work in the field of imbalanced classification. Coding is done by altering, rewriting, and cleaning up parts of the original LNR paper’s GitHub repository.

Both a binary classification using KEEL datasets and a multiclass classification using CIFAR-10 are reproduced successfully. Results closely match those reported in the original paper, confirming a correct reimplementaion of the LNR method.

10.2 Grades D-C-B

Grade C requires additional empirical experiments to be documented. This was done by tuning the LNR hyperparameter t_{flip} on the Meter2800 dataset, showing its effect on minority class performance. Meters 5/4 and 7/4 are rare in Western music, making their detection challenging. Observing the resulting macro F1-scores and accuracies for different t_{flip} values, we found that a moderate flip threshold of $t_{flip} = (SENARE)$ yielded the best balance between proper minority class detection and performance.

10.3 Grade B-A

Going beyond the scope of the original paper, we applied the LNR method to do time signature detection on the imbalanced Meter2800 dataset. This novel application required adapting the LNR algorithm to a different domain and data modality. Audio files were altered to be visualized as MFCC images, suitable for a tailored one-channel ResNet18 input.

Differing thresholds t_{flip} were also tested to find the optimal setting for this task. Doing so helped bring a theoretical method into practical use, along with the team’s further examples of real-world applications.

References

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